

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	06 July 2026
Team ID	
Project Name	Ledgerline – Credit Card Approval Prediction System
Maximum Marks	4 Marks

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Applicant Data Intake	Single-applicant web form • Input validation for all fields • JSON API endpoint for programmatic access
FR-2	Data Preprocessing & Feature Engineering	Merge application and credit history records • Convert multi-class STATUS codes to binary risk label (60+ days past due = high risk) • Derive AGE_YEARS, YEARS_EMPLOYED, INCOME_PER_FAMILY_MEMBER • One-hot encode categorical features
FR-3	Model Training & Selection	Train Logistic Regression, Decision Tree, Random Forest, and XGBoost • Compare all four via Accuracy, Precision, Recall, F1, and ROC-AUC • Auto-select the best model by ROC-AUC • Calibrate decision threshold to maximize F1
FR-4	Prediction & Decisioning	Return Approve/Decline decision • Return calibrated confidence score • Display key contributing factors
FR-5	Batch Screening	CSV upload for multiple applicants • Bulk scoring using the same model as single-applicant flow • Downloadable results CSV • Summary totals (approved/declined count)
FR-6	Model Transparency Report	Model comparison bar chart • ROC curves for all four models • Feature importance / coefficient chart for the selected model

Non-functional Requirements: Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple, form-based UI requiring no ML knowledge from analysts, compliance staff, or customers. Clear Approve/Decline result with confidence score and plain-language details.
NFR-2	Security	Server-side input validation on all form and API fields. Model artifacts and credentials are never exposed to the client. No hard-coded API keys or secrets in the codebase.

NFR-3	Reliability	Deterministic preprocessing pipeline with a fixed feature column order (feature_columns.json) ensures the same input always produces the same prediction, regardless of when it is run.
NFR-4	Performance	Single-applicant predictions return in well under 1 second (model loaded once at app startup). Batch screening of dozens of applicants completes in a few seconds.
NFR-5	Availability	Designed to run as a standalone Flask service. Can be containerized for continuous availability. Optional cloud deployment via IBM Watson Machine Learning for always-on access.
NFR-6	Scalability	Stateless prediction logic (model and scaler loaded once at startup) allows horizontal scaling behind a load balancer. IBM Watson Machine Learning deployment supports scaling to high request volumes without changing the application code.